

Chapter: Blackbody

Problems

1. Sunspots appear dark, although their temperature are typically 5000 K because the rest of the sun's surface is even hotter, about 5800 K. Compare the radiation rates of surfaces of the same emissivity whose temperatures are respectively 5000 and 5800 K.
2. At what rate would solar energy arrive at the earth if the solar surface had a temperature 10 percent lower than it is?
3. An object is at a temperature of 400 °C. At what temperature would it radiate energy twice as fast?
4. Find the surface area of a blackbody that radiates 1.00 kW when the temperature is 500 °C. If the blackbody is a sphere, what is its radius?
5. Find the peak wavelength in the spectrum of the radiation from a blackbody at a temperature of 500 °C. In what part of the electromagnetic spectrum is this wavelength?
6. The brightest part of the spectrum of the star Sirius is located at a wavelength of about 290 nm. What is the surface temperature of Sirius?

Chapter: Photo-Electric Effect

Problem Set

1. The maximum wavelength for photoelectric emission in tungsten is 230 nm. What wavelength of light must be used in order for electrons with maximum energy of 1.5 eV to be ejected?
2. The minimum frequency for photoelectric emission in copper is 1.1×10^{15} Hz. Find the maximum energy of the photoelectrons (in electronvolts) when light of frequency 1.5×10^{15} Hz is directed on a copper surface?
3. What is the maximum wavelength of light that will cause photoelectrons to be emitted from sodium? What will the maximum kinetic energy of the photoelectrons be if 200-nm light falls on a sodium surface? (work function for sodium is 2.3 eV).
4. A metal surface illuminated by 8.5×10^{14} Hz light emits electrons whose maximum energy is 0.52 eV. The same surface illuminated by 12.0×10^{14} Hz light emits electrons whose maximum energy is 1.97 eV. From these data find Planck's constant and the work function of the surface.
5. The work function of a tungsten surface is 5.4 eV. When the surface is illuminated by light of wavelength 175 nm, the maximum photoelectron energy is 1.7 eV. Find the Planck's constant from these data.

Chapter: Compton Effect

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1. A monochromatic x-ray beam whose wavelength is 55.8 pm is scattered through 46° . Find the wavelength of the scattered beam.
 2. A beam of x-rays is scattered by a target. At 45° from the beam direction the scattered x-rays have a wavelength of 2.2 pm. What is the wavelength of the x-rays in the direct beam?
 3. An x-ray photon whose initial frequency was $1.5 \times 10^{19} \text{ Hz}$ emerges from a collision with an electron with frequency of $1.2 \times 10^{19} \text{ Hz}$. How much kinetic energy was imparted to the electron?
 4. An x-ray photon of initial frequency $3.0 \times 10^{19} \text{ Hz}$ collides with an electron and is scattered through 90° . Find its new frequency.
 5. At what scattering angle will incident 100-keV x-rays leave a target with an energy of 90 keV?

Chapter: X-ray

1. What voltage must be applied to an x-ray tube for it to emit x-rays with a minimum wavelength of 30 pm?
2. Electrons are accelerated in television tubes through potential differences of about 10 kV. Find the highest frequency of the electromagnetic waves emitted when these electrons strike the screen of the tube. What kind of waves are these?